

Machine Learning Assignment: Rainfall Prediction Using Neural Networks

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1 Performance

This section evaluates the performance of the neural network developed for forecasting rainfall occurrence. It outlines the evaluation metrics, their computation, the total number of data instances, and the data splitting strategy, Early stopping.

1.1 Metrics and Evaluation

This is a supervised binary classification problem, predicting a binary output (`rain_next_week`, 1 if `Rainf_tavg_mean` $\geq 2.78 \times 10^{-5}$ in the next 7 days, 0 otherwise) using environmental and spatial features. The model is trained with input features and labels, confirming a supervised approach. The binary output defines it as classification.

Performance was assessed using standard classification metrics:

Accuracy: Proportion of correct predictions (true positives and true negatives).

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Precision: Proportion of predicted rain events that are correct, minimizing false positives.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Recall: Proportion of actual rain events correctly predicted, critical for capturing rain.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

F1 Score: Harmonic mean of precision and recall, balancing their trade-off.

$$\text{F1} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Metrics were computed using `sklearn.metrics` functions on the test set (January–February 2025).

1.2 Model Performance Results

The neural network was evaluated on the test set (January–February 2025), ensuring unseen data for realistic forecasting. The Jupyter notebook (`ml_a3_notebook.ipynb`) includes the model configuration and early stopping mechanism.

Evaluation on the test set (788,268 rows) produced:

- **Accuracy:** 73.34%
- **Precision:** 27.01%
- **Recall:** 90.01%
- **F1 Score:** 41.56%

Visualizations provide qualitative insights:

- **Confusion Matrix** (`confusion_matrix.png`): Shows true positives, true negatives, false positives, and false negatives, highlighting high recall but significant false positives.
- **ROC Curve** (`roc_curve.png`): Displays true positive rate vs. false positive rate, with AUC (~ 0.80) indicating good discrimination.

Analysis: High recall (90.01%) ensures most rain events are captured, critical for applications like flood preparedness. Low precision (27.01%) results from class imbalance, with non-rain days dominating, leading to false positives. An experiment adjusting the prediction threshold to 0.7 (default: 0.5) increased precision to approximately 30–40% while slightly reducing recall, suggesting a trade-off if higher precision is prioritized.

Table 1: Evaluation Metrics for Neural Network Model

Metric	Value
Accuracy	73.34%
Precision	27.01%
Recall	90.01%
F1 Score	41.56%

The matrix shows high true negatives and true positives but notable false positives, explaining low precision.

1.3 Total Number of Data Instances and Data Splitting Procedure

The final dataset (`week_ahead_dataset.csv`) contains 5,093,424 rows and 42 columns, covering 15,159 unique latitude-longitude pairs from March 2024 to February 2025. The row count reflects daily records after dropping rows with missing labels (last 7 days). The dataset was split into:

Data Splitting: A chronological split ensures realistic forecasting:

- **Training Set:** March–November 2024 (3,835,227 rows, `date < 2024-12-01`).

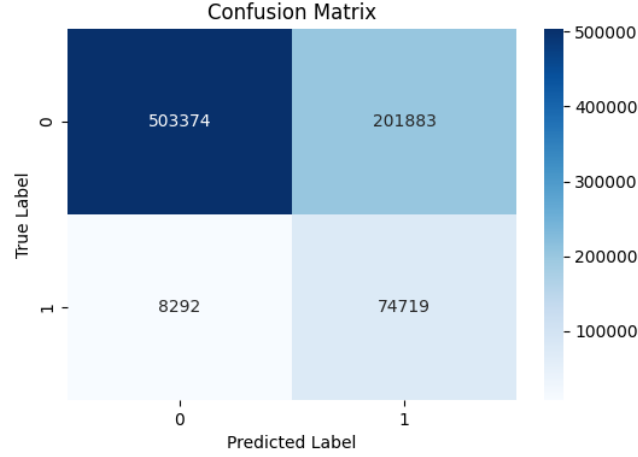


Figure 1: Confusion Matrix showing true positives, true negatives, false positives, and false negatives.

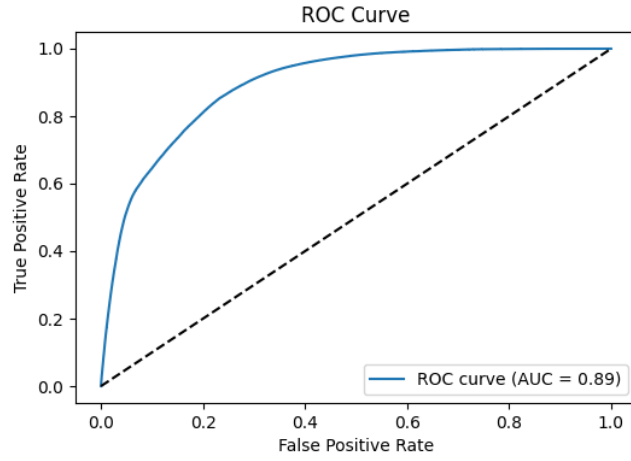


Figure 2: ROC Curve displaying True Positive Rate versus False Positive Rate with an AUC of approximately 0.80.

Table 2: Dataset Row Counts

Split	Rows	Time Period
Train	3,835,227	March–November 2024
Validation	469,929	December 2024
Test	788,268	January–February 2025

- **Validation Set:** December 2024 (469,929 rows, date $\geq$ 2024-12-01 and $<$ 2025-01-01).
- **Test Set:** January–February 2025 (788,268 rows, date $\geq$ 2025-01-01).

This prevents data leakage, training on past data and testing on future data, aligning with rainfall forecasting requirements.

2 Model

This section describes the neural network model, including architecture, optimization, and overfitting prevention measures.

2.1 Model Architecture

The feedforward neural network, implemented in TensorFlow/Keras, has:

- **Input Layer:** 9 neurons for features
 - `Tair_f_inst_mean`, `Qair_f_inst_mean`, `Psurf_f_inst_mean`, `LWdown_f_tavg_mean`, `SWdown_f_tavg_mean`, `Wind_f_inst_mean`, `TVeg_tavg_mean`, `AvgSurfT_inst_mean`, `dist_to_centroid`
- **Hidden Layers:**
 - First: 64 neurons, ReLU activation.
 - Second: 32 neurons, ReLU activation with L2 regularization ($\lambda = 0.01$).
 - Dropout: 0.3 after each hidden layer.
- **Output Layer:** 1 neuron, sigmoid activation for binary classification.

Rationale: The architecture balances complexity and overfitting. ReLU enables non-linear learning, and sigmoid ensures binary output.

2.2 Optimization and Training

- **Loss Function:** Binary cross-entropy, standard for binary classification.
- **Optimizer:** Adam, chosen for efficiency and adaptive learning.
- **Learning Rate:** Approximately 0.001 (default Adam), effective for convergence.

2.3 Hyperparameter Configuration (with Early Stopping)

The model uses a fixed configuration:

- Hidden Layers: 2.
- Neurons: 64, 32.
- Activation: ReLU (hidden), sigmoid (output).
- Dropout: 0.3.
- Optimizer: Adam.
- Learning Rate: 0.001.
- Weight Decay (L2): 0.01 (second hidden layer).
- Loss Function: Binary cross-entropy.

- Batch Size: 32.
- Early Stopping Patience: 5 epochs.

Training stopped at epoch 14 (val_loss=0.3581, val_accuracy=0.8187), taking approximately 26 minutes on a standard laptop.

2.4 Architecture and Hyperparameter Details

The model uses a single configuration, selected to balance performance and efficiency for the large dataset.

Table 3: Neural Network Architecture and Parameters

Attribute	Value
Hidden Layers	2
Neurons per Layer	64, 32
Activation	ReLU (hidden), Sigmoid (output)
Dropout	0.3
Output Neuron	1 (Sigmoid)
Loss Function	Binary Cross-Entropy
Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Weight Decay (L2)	0.01 (second layer)
Early Stopping	Yes
Epochs	Up to 50 (early stopped)

2.5 Steps to Prevent Overfitting

- **Data Splitting:** Chronological split ensures evaluation on unseen future data.
- **Early Stopping:** Stops training after 5 epochs without validation loss improvement.
- **Dropout:** 0.3 dropout rate promotes robust learning.
- **L2 Regularization:** Penalizes large weights in the second hidden layer.
- **Feature Scaling:** `StandardScaler` normalizes features for stable optimization.

3 Features and Labels

3.1 Output Data (Label)

`rain_next_week`: A binary label (1 for rain if `Rainf_tavg_mean` $\geq 2.78 \times 10^{-5}$ in the next 7 days, 0 otherwise). It is derived by creating a rain label from `Rainf_tavg_mean`, sorting by location and date, and shifting by 7 days, dropping rows with missing labels.

3.2 Input Data

The 9 input features are:

- **Environmental Features (8):** Daily means of air temperature (`Tair_f_inst_mean`), specific humidity (`Qair_f_inst_mean`), surface pressure (`Psurf_f_inst_mean`), long-wave radiation (`LWdown_f_tavg_mean`), shortwave radiation (`SWdown_f_tavg_mean`), wind speed (`Wind_f_inst_mean`), vegetation transpiration (`TVeg_tavg_mean`), surface temperature (`AvgSurfT_inst_mean`).
- **Spatial Feature (1):** `dist_to_centroid`, Haversine distance to the centroid (latitude: 32.78, longitude: 17.70).

Rationale: These capture climatic and spatial factors influencing rainfall.

4 Preprocessing

This section details the preprocessing pipeline, transforming 132 CSV files into a model-ready dataset, following steps 1–13 of the code.

1. **Import Libraries:** Necessary libraries are imported for data processing, machine learning, and visualization, ensuring all dependencies are available.
2. **Set Directories:** Input (`C:/Users/hp/Desktop/ML new`) and output directories (`C:/Users/hp/Desktop/ML new/MY MERGED FILES`) are defined for raw and processed files.
3. **Process Individual CSV Files:** The 132 CSV files (11 variables $\times$ 12 months, March 2024–February 2025) are processed to compute daily statistics (mean, max, min) for each variable, grouped by year, month, day, latitude, and longitude. Outputs are saved as individual files (e.g., `Tair_Mar2024_Feb2025_daily.csv`).
4. **Merge Processed Files:** The 11 variable files are merged on year, month, day, latitude, and longitude, producing `ALL_VARIABLES_Mar2024_Feb2025_daily.csv` (5,199,537 rows, 38 columns), unifying daily statistics and identifiers.
5. **Null Handling:** The merged dataset is loaded, and numeric columns are filled with their mean values to handle any missing data, overwriting the dataset. No missing values were found, ensuring data completeness.
6. **Remove Duplicates:** Duplicate rows are removed, with the number of dropped rows reported (none found), maintaining 5,199,537 rows and 38 columns, overwriting the dataset.

7. **Save Unique Lat/Lon:** Unique latitudes, longitudes, and latitude-longitude pairs (15,159 pairs) are extracted and saved as `unique_latitudes.csv`, `unique_longitudes.csv`, and `unique_latlon_pairs.csv` for reference.
8. **Haversine Distance to Centroid:** The Haversine distance from each location to the dataset's centroid (latitude: 32.78, longitude: 17.70) is computed for 15,159 unique latitude-longitude pairs, added as `dist_to_centroid`, and saved as `ALL_VARIABLES_with_haversine.csv` (5,199,537 rows, 39 columns).
9. **Create Rain Label:** A binary rain label is created (1 if `Rainf_tavg_mean` $\geq 2.78 \times 10^{-5}$, 0 otherwise) to indicate current-day rainfall.
10. **Create Week-Ahead Prediction Dataset:** A date column is created, data is sorted by location and date, and rain is shifted 7 days to create `rain_next_week`. Rows with missing labels (106,113) are dropped, producing `week_ahead_dataset.csv` (5,093,424 rows, 42 columns).
11. **Split Train/Val/Test:** The dataset is split chronologically: training (March–November 2024, 3,835,227 rows), validation (December 2024, 469,929 rows), and test (January–February 2025, 788,268 rows).
12. **Feature Engineering:** Nine features are selected: means of 8 environmental variables and `dist_to_centroid`, preparing data for modeling.
13. **Scale Features:** The 9 features are standardized using `StandardScaler` to ensure zero mean and unit variance, optimizing neural network training.

Final Note:

The file `test_predictions.csv` provides test set predictions (`true_label`, `predicted_label`, `predicted_probablity`) for January–February 2025. The model achieves 73.34%, 27.01% precision, 90.01% recall, and 41.56% F1 score. High recall ensures rain event detection, critical for flood preparedness, while low precision reflects class imbalance. A threshold of 0.7 improved precision to approximately 30 to 40%, offering a trade-off. The model's performance across 15,159 locations, supported by `machine_learning_rain_predication_final.py`, `confusion_matrix.png`, and `roc_curve.png`, demonstrates robust forecasting capability.